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Lifestyle factors and jet lag prevention: a cross-sectional study of travel wellness among Japanese and U.S. business class travelers

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Abstract

Purpose

We intend to examine whether behavioral and mental health-related determinants including sleep, stress level, smoking, drinking, physical activity, and vegetable intake were associated with severity of jet lag and symptoms of jet lag.

Methods

We delivered a survey to ANA HOLDINGS (ANA) customers in Japan and U.S. asking for their lifestyle information, subjective jet lag perceptions and symptoms of jet lag, and obtained a sample of 1,759 Japanese and 483 U.S. participants.

Results

Multivariate linear regression analysis showed a positive association between severity of jet lag perception and perceived stress (Model 2: [b=0.29, 95%CI: 0.10; 0.48]), but a negative association between severer jet lag perception and both smoking (Model 2: [b=-0.20, 95%CI: -0.35; -0.05]) and physical activity (Model 2: [b=-0.13, 95%CI: -0.25; -0.004]). Results revealed a positive association between higher jet lag symptoms and perceived stress (model2: [b=0.39, 95%CI: 0.26; 0.53]), daytime sleepiness (Model 2: [b=0.02, 95%CI: 0.003; 0.04]), and vegetable consumption (Model 2: [b=0.10, 95%CI: 0.03; 0.18]).

Conclusions

Daytime sleepiness and stress were associated with jet lag. Results varied for other lifestyle factors including smoking, physical activity, vegetable intake, and alcohol. Given the health risks among business travelers, it is important to develop evidence-based jet lag prevention strategies.

Key words: jet lag, lifestyle, behavior change, stress, sleep

Introduction

Despite advances in technology, businesses still place great value on face-to-face meetings. More than 405 million American people make long-distance business travel every year: the 16% of them accounts for long-distance travel [1]. In fact, the number of business travelers has been increasing globally. In 2018, traveling for business and professional purpose account for 13% of all international travels [2], and this tendency has been predicted to increase overtime [3]. With the rise of these long distance business travel, business travelers face with more health risks than leisure travelers [4] . These adverse health outcomes include symptoms of depression and anxiety and sleeping problems [5]. While business trips offer positive opportunities such as job benefits and occupational training advancement for employees [6], the health risks associated with frequent travel may offset these benefits.

Jet lag is one consequence of travel that may pose a health risk. Jet lag, also known as circadian desynchrony, is associated with increased risk of insomnia [7]. Repeated long-haul flights also increase risk of potential cognitive deficits or psychotic and major affective disorders [8]. In the long run, sleep and circadian rhythm disruption are reported to be associated with a broad range of interconnected pathologies such as reduced motivation, depression, metabolic abnormalities, obesity, immune impairment, and even a greater risk of cancer [9], [10].

Due to the mismatch between internal body clock and external environment caused by rapidly crossing three, four, or more time zones, jet lag brings poor sleep, increased travel fatigue, headaches, lack of focus, and gestational disturbance. Unlike travel fatigue, which can disappear within a day or so, jet lag can persist for longer periods of time. The number of days until recovery can equal roughly two-thirds the number of time zones crossed, and about half the number of time zones crossed after westward flights [8]. Individual recovery time varies.

Researchers have explored ways to manage jet lag, through both non-pharmacological and pharmacological means. However, despite the known short- and long-term health risks, surprisingly few studies have focused on this topic. One study found that jet lag is potentially associated with lifestyle such as sleep deprivation, stress, and excessive caffeine and alcohol intake [8], [11]. Previous research has identified a strong association between frequency of business travel and physical and behavioral risks, including poor self-rated health; clinical symptoms of anxiety and depression; alcohol consumption and smoking; a lack of physical activity; and sleep problems [5]. However, to our knowledge, there is a gap in the literature exploring lifestyle determinants of jet lag beyond assessment of individual variation. Most of the existing research in this area -

particularly related to interventions - has targeted athletes and flight crews [12]–[16]. Jet lag among business travelers remains understudied, and there is a need for more rigorous scientific analysis in this area.

To address these critical gaps, our study investigates the relationship between lifestyle factors (i.e., health behaviors) and jet lag among a sample of business travelers. In particular, we examine whether sleep, stress level, smoking, drinking, physical activity, and vegetable intake are associated with severity of jet lag. We also test whether these lifestyle factors are related to jet lag symptoms. We hypothesize that people with healthier lifestyles (no sleep issues, low stress, no smoking, higher vegetable intake, no alcohol consumption) have a less severe perception of jet lag and report fewer jet lag symptoms.

METHODS

Study design

This is a cross-sectional study investigating the associations between jet lag and health status/lifestyle among Japan and U.S. business class travelers.

Participants

We conducted this survey among ANA users who met the following inclusion criteria: 1) membership in ANA mileage club (frequent flier program), 2) those with business class flight experience (including other airlines) within the past year, and 3) those living in Japan or the U.S. People were offered via internet to participate in the survey with an incentive of ANA mileage point (200 mileage points: although the value of mileage point is not determined uniformly, it is exchanged for 10,000 mileage points equivalent to approximately 100 USD when exchanging miles with an electronic money of a partner company. According to this calculation, the authors assume that 200 mileage points, the incentive of the present survey is equivalent to approximately 2 USD). The participants who agreed to take survey answered each question through a PC or smartphone. For participants living in Japan, the Japanese version of the survey was provided. For those living in U.S., the English version of the survey was provided. Ultimately, we collected 1,759 Japanese and 484 English completed surveys. Researcher received survey data as an anonymous csv file from surveyor (ANA).

Measures

Dependent variable

Jet lag perception

Jet lag perception refers to subjective severity of jet lag in general noted by the participants. We measured jet lag perception with a question asking "on a scale of 1 to 10, how would you rate the severity of your jet lag overall?" using Visual Analog Scale (VAS) ranging from "insignificant jet lag" (1) to "very bad jet lag" (10) based on the Liverpool jet lag scale [17]. Participants' answers were coded as continuous variable ranging from 1 to 10. We translated the scale from its original English into Japanese. Although the scale in Japanese has not been validated, the translated version was discussed by authors fluent in both English and Japanese until a consensus was reached.

Jet lag symptoms

Jet lag symptoms refer to the number of jet lag-related symptoms in general participants reported. We measured jet lag symptoms with 11 questions: delayed sleep onset, early awakening, fractionated sleep, poor performance during the new daytime at physical tasks, poor performance during the new daytime at mental tasks, increased fatigue, frequency of headaches and irritability, decreased ability to concentrate, nausea or lack of appetite, constipation or other irregularity of bowel movements, and "other." We selected these symptoms based on the previous study explaining the jet lag symptoms [8]. We summed the number of symptoms participants reported and used the score as continuous variable representing the degree of jet lag symptom.

Independent variables

Perceived stress

Perceived stress was measured using the Perceived Stress Scale (PSS) proposed by Cohen [18]. PSS is the most widely used psychological instrument for measuring the general perception of stress. It is a measure of the degree to which situations in one's life are appraised as stressful. We captured perceived stress through 10 questions: "1. In the last month, how often have you been upset because of something that happened unexpectedly?", "2. In the last month, how often have you felt that you were unable to control the important things in your life?", "3. In the last month, how often have you felt nervous and 'stressed'?", "4. In the last month, how often have you felt confident about your ability to handle your personal problems?", "5. In the last month, how often have you felt that things were going your way?", "6. In the last month, how often have you found that you could not cope with all the things that you had to do?", "7. In the last

month, how often have you been able to control irritations in your life?", "8. In the last month, how often have you felt that you were on top of things?", "9. In the last month, how often have you been angered because of things that were outside of your control?" and "10. In the last month, how often have you felt difficulties were piling up so high that you could not overcome them?" Each question was asked using a 5-point Likert scale ranging from (1) "Never" to (5) "Very often." PSS scores were obtained by reverse-coding responses for the four positively stated items (items 4, 5, 7, and 8) and then summing across all scale items. We used PSS scores as continuous variable.

Daytime sleepiness

Daytime sleepiness was measured using the Epworth Sleepiness Scale (ESS) proposed by Johns [19]. ESS is a scale intended to measure daytime sleepiness and is used for diagnosing sleep disorders. The questionnaire asks the subject to rate his or her probability of falling asleep on a scale of increasing probability from 0 to 3 during eight different situations that most people engage in daily, including: "sitting and reading," "watching TV," "sitting, inactive in a public place (e.g., a theatre or a meeting)," "as a passenger in a car for an hour without a break," "lying down to rest in the afternoon when circumstances permit," "sitting quietly after a lunch without alcohol," and "in a car, while stopped for a few minutes in traffic." The scores for the eight questions are added together to obtain a single continuous variable ranging from 0 to 24. A value in the 0-9 range is considered normal, while a number in the 10-24 range indicates that expert medical service should be sought.

Behavioral risk factors

The lifestyle of participants was measured using questions from the Behavioral Risk Factor Surveillance System (BRFSS) [20]. Run by the Centers for Disease Control and Prevention (CDC), BRFSS is the largest continuous health survey in the US examining behavioral risk factors. Each year data are collected in all 50 states as well as the District of Columbia and three U.S. territories.

Smoking

Smoking status was measured with the question "Do you currently smoke cigarettes? (tobacco and/or e-cigarettes)" Participants chose an answer from "Yes," "No, but I used to smoke" or "No, I have never smoked." We coded "Yes" as 3, "No, but I used to smoke" as 2, and "No, I have never smoked" as 1. We used it as continuous variable.

Drinking

Drinking status was measured with the question "Do you currently drink alcohol?" Participants chose an answer from "Yes," "No, but I used to drink" or "No, I never drink alcohol." We coded "Yes" as 3, "No, but I used to drink" as 2, and "No, I never drink alcohol" as 1. We used it as continuous variable.

Vegetable consumption

Vegetable consumption was measured with the question "How often did you eat fruits and vegetables over the past month?" Participants chose an answer from "Twice a day or more," "Once a day," "Four or six times a week," "Once a week," "Less than once a week," or "None." We coded "Twice a day or more" as 6, "Once a day" as 5, "Four or six times a week" as 4, "Once a week" as 3, "Less than once a week" as 2, and "None" as 1. We used it as continuous variable.

Physical activity

Physical activity was measured with the question "During the past month, other than your regular job, did you participate in any physical activities or exercises such as running, calisthenics, golf, gardening, or walking for exercise?" Participants chose an answer from "Yes," "No," or "Don't know/Not sure." We coded "Yes" as 1, and "No" and "Don't know/Not sure" as 0. Thus we used it as binary variable.

Covariates

Age, sex, employment, income, education, marital status, the purpose of flight, and survey language (Japanese or English) were used as covariates.

Statistical analysis

We performed descriptive analyses to summarize the characteristics of our study sample (Table 1) and of our dependent variables (jet lag perception and jet lag symptom) (Figure 1 and Figure 2). To assess the associations between dependent variables and health status/lifestyle, we ran multivariate linear regressions. The first analysis treated jet lag perception as the dependent variable, while the second analysis treated jet lag symptom as the dependent variable. For each analysis, we ran two models: in Model 1, we adjusted for gender, age, the purpose of flight, survey language, and health status/lifestyle. In Model 2, we further adjusted for socio-economic status (occupation, income, education, and marital status). P-values below 0.05 were considered statistically significant. Statistical analyses were performed using R version 3.3.2 [21].

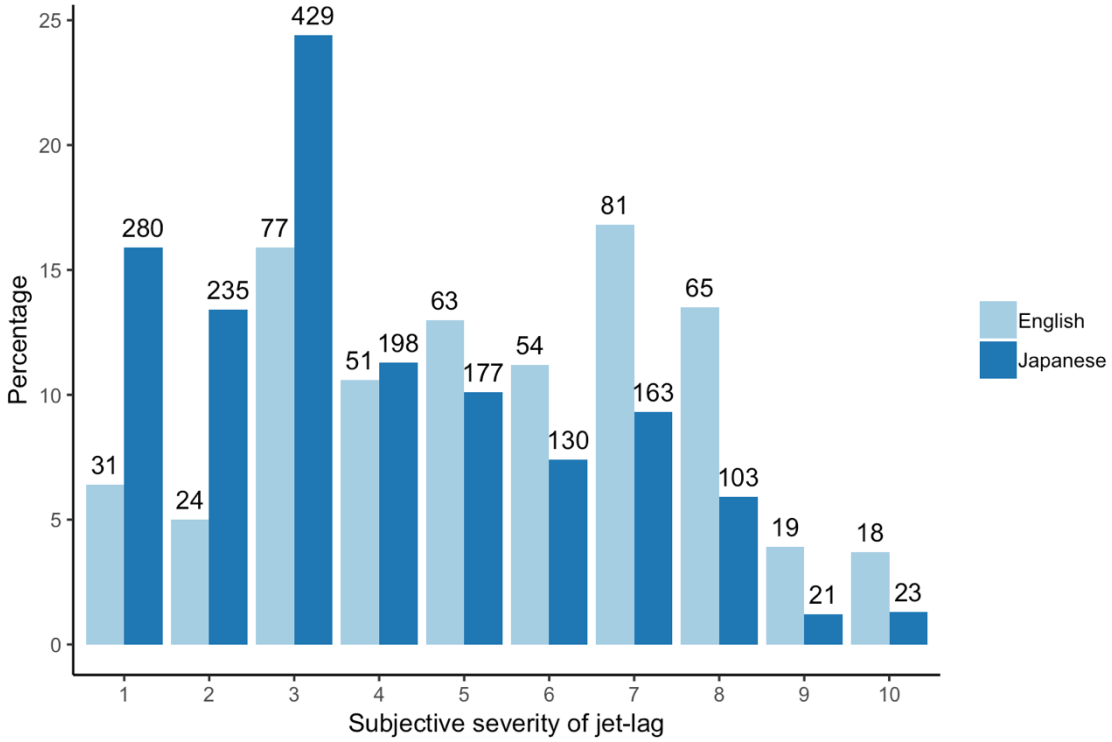

Figure 1. Distribution of jet lag perception for Japanese and English participants.

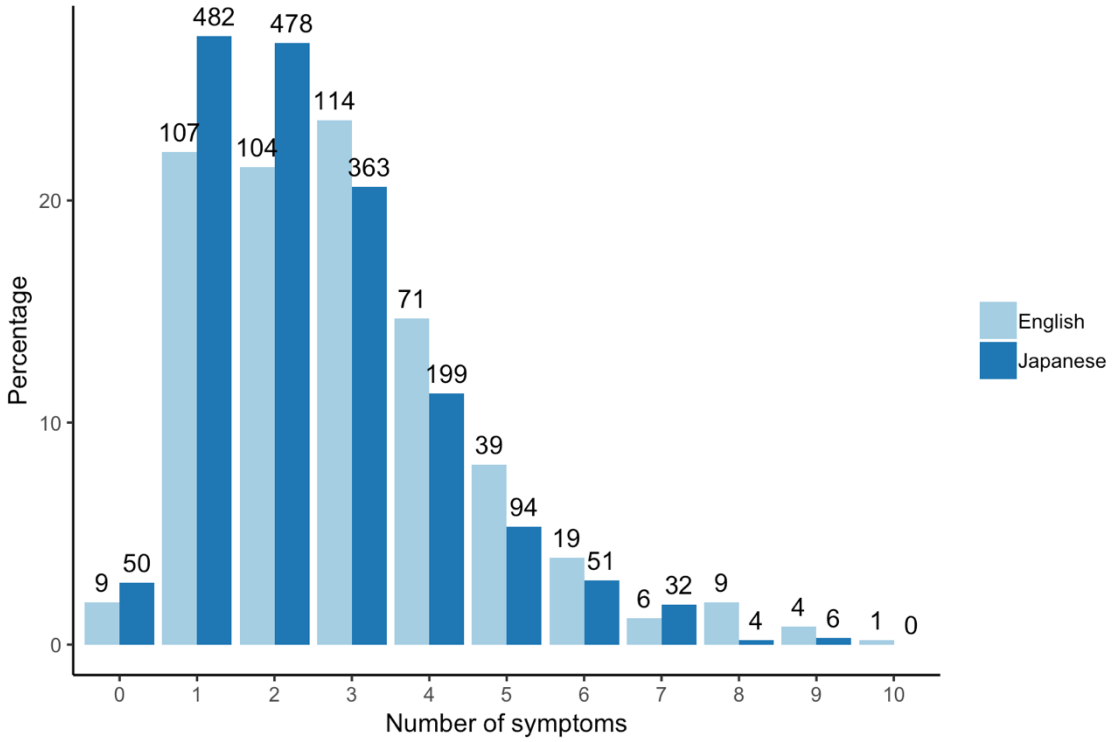

Figure 2. Distribution of jet lag symptoms for Japanese and English participants.

Results

Demographic statistics of the study sample are shown in Table 1. Our sample included 1,759 Japanese and 483 English participants. Most participants were male, accounting for nearly 70-90% of the sample. The mean age was 50.5 and 50.6 for Japanese and English version, respectively. Fifty to sixty percent of participants reported being employed. Median income was 115,000 and 165,000 USD for Japanese and English participants, respectively. In terms of educational attainment, Japanese respondents showed the highest percentage (52.1%) of "university," while U.S. respondents showed the highest percentage (43.7%) of "graduate school." Roughly 65-75% of participants reported being "married" or in a "domestic partnership." Nearly 30% of both language groups reported living alone. With regard to purpose of flight, more Japanese respondents reporting flying for "business," whereas more U.S. respondents reported flying for "leisure," although answers for most of the participants in both groups had missing values.

Table 1. Demographic characteristics of the study sample

Variables	Japanese (n=1,759)	English (n=483)	P-value
Gender:			
Female	255 (14.5)	141 (29.2)	
Male	1,504 (85.5)	342 (70.8)	
Mean age (mean, SD):	50.5 (9.6)	50.6 (13.1)	
Age group (years):			
10-19	0 (0.0)	4 (0.8)	
20-29	29 (1.6)	30 (6.2)	
30-39	205 (11.7)	57 (11.8)	
40-49	535 (30.4)	119 (24.6)	<0.001
50-59	679 (38.6)	144 (29.8)	
60-69	280 (15.9)	101 (20.9)	
70-79	31 (1.8)	25 (5.2)	
80-89	0 (0.0)	3 (0.6)	
Employment:			
Self-employed	243 (13.8)	129 (26.7)	<0.001

Employed	1,216 (69.1)	258 (53.4)	
Not employed	5 (0.3)	3 (0.6)	
Homemaker	56 (3.2)	20 (4.1)	
Student	6 (0.3)	10 (2.1)	
Retired	42 (2.4)	37 (7.7)	
Missing	190 (10.8)	26 (5.4)	
Median income (Median, IQR)	115,000 (100,000)	165,000 (145,000)	
Income(USD/year)			
<30,000	14 (0.8)	6 (1.2)	
30,000-49,999	46 (2.6)	21 (4.3)	
50,000-69,999	106 (6.0)	21 (4.3)	
70,000-89,999	211 (12.0)	19 (3.9)	
90,000-109,999	236 (13.4)	44 (9.1)	
110,000-129,999	228 (13.0)	37 (7.7)	
130,000-149,999	155 (8.8)	33 (6.8)	
150,000-169,999	88 (5.0)	28 (5.8)	<0.001
170,000-189,999	73 (4.2)	19 (3.9)	
190,000-209,999	114 (6.5)	26 (5.4)	
210,000-229,999	39 (2.2)	11 (2.3)	
230,000-249,999	33 (1.9)	33 (6.8)	
>250,000	200 (11.4)	141 (29.2)	
Missing	216 (12.3)	44 (9.1)	
Education:			
Junior high school	2 (0.1)	3 (0.6)	
High school	132 (7.5)	15 (3.1)	
College or technical school	134 (7.6)	43 (8.9)	
University	917 (52.1)	183 (37.9)	<0.001
Graduate school	388 (22.0)	211 (43.7)	
Unknown	186 (10.6)	28 (5.8)	
Marital status:			
Single, never married	211 (12.0)	78 (16.1)	
Married or domestic partnership	1,190 (67.7)	351 (72.6)	<0.001
Widowed	19 (1.1)	6 (1.2)	
Divorced	60 (3.4)	15 (3.1)	

Separated	89 (5.1)	4 (0.8)	
Missing	190 (10.8)	29 (6.0)	
Household size:			
1	572 (32.5)	158 (32.7)	
2	441 (25.1)	154 (31.9)	
3	341 (19.4)	83 (17.2)	0.007
4	331 (18.8)	65 (13.5)	
>5	74 (4.2)	23 (4.8)	
Purpose of flight:			
Business	439 (25.0)	92 (19.0)	
Leisure	59 (3.4)	118 (24.4)	
Meeting someone	1 (0.1)	6 (1.2)	<0.001
Private event	6 (0.3)	1 (0.2)	
Missing	1254 (71.3)	266 (55.1)	

Values are numbers and percentages unless otherwise stated.
Chi-square tests were performed for each comparison.

Descriptive statistics of jet lag perception and jet lag symptoms are shown in Figure 1 and Figure 2. Among jet lag perception categories, the highest percentage of Japanese respondents fell into the 3 grouping, whereas U.S. respondents showed the highest percentage in "7." In terms of jet lag symptoms, the highest percentage of Japanese respondents reported "1," while U.S. respondents most frequently reported "3."

The results of multivariate linear regression analysis using jet lag perception as the dependent variable are shown in Figure 3. For all independent variables, estimated coefficients represent how much the jet lag perception increases as health status (perceived stress and daytime sleepiness) or behavior (smoking, drinking, vegetable consumption, and physical activity) increase. Perceived stress was positively associated with higher jet lag perception in Model 2 [b=0.31, 95%CI: 0.12; 0.50]. On the other hand, smoking was negatively associated with higher jet lag perception in Model 1 [b=-0.26, 95%CI:-0.40; -0.12] and Model 2 [b=-0.21, 95%CI:-0.36; -0.06] (See supplementary Table 1 for full model).

The results of multivariate linear regression analysis using jet lag symptoms as the dependent variable are shown in Figure 4. For all independent variables, estimated

coefficients represent how much the jet lag symptoms increases as health status (perceived stress and daytime sleepiness) or behavior (smoking, drinking, vegetable consumption, and physical activity) increase. Perceived stress was positively associated with higher numbers of jet lag symptoms in Model 1 [b=0.17, 95%CI: 0.08; 0.26] and in Model 2 [b=0.40, 95%CI: 0.27; 0.54]. Also, daytime sleepiness was positively associated with higher numbers of jet lag symptoms in Model 1 [b=0.03, 95%CI: 0.01; 0.05] and Model 2 [b=0.02, 95%CI: 0.004; 0.04]. While smoking was negatively associated with the number of jet lag symptoms in Model 1 [b=-0.11, 95%CI: -0.21; -0.01], the association was attenuated in Model 2. Moreover, vegetable consumption was positively associated with the number of jet lag symptoms in Model 1 [b=0.12, 95%CI: 0.05; 0.20] and model 2 [b=0.11, 95%CI: 0.03; 0.18] (See Supplementary Table 2 for full model).

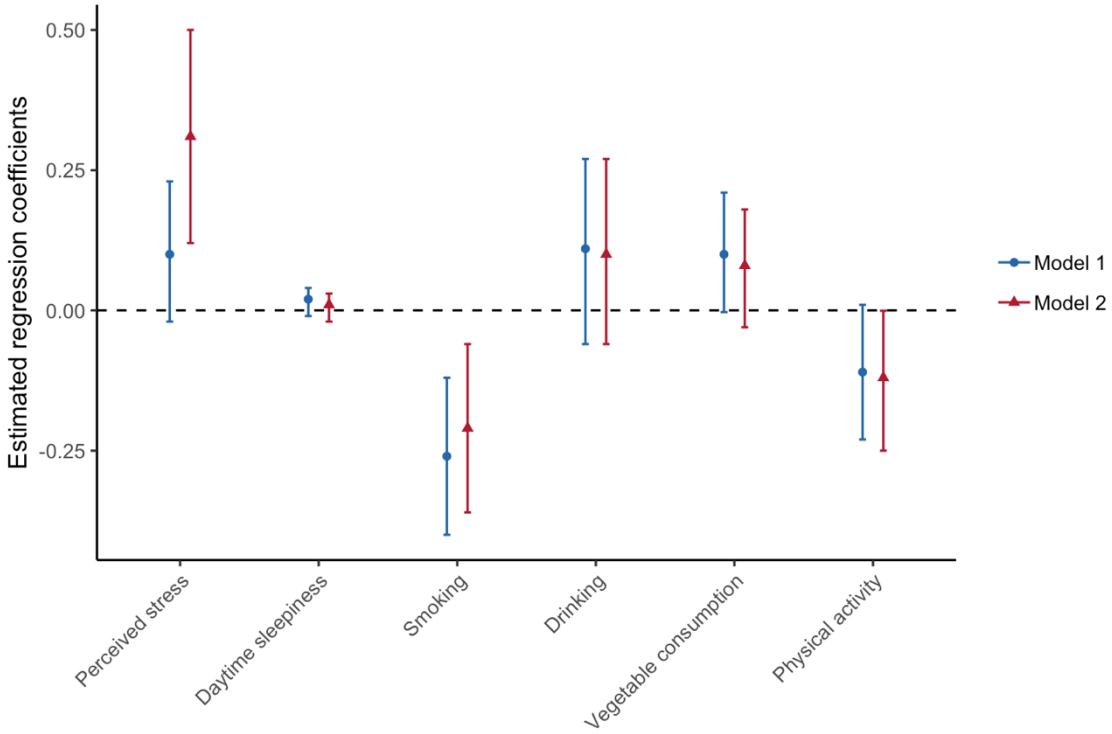

Figure 3. Estimated regression coefficients of jet lag perception on health status / behavior. Model 1: Adjusted for gender, age, the purpose of flight, language and other health behavior. Model 2: Adjusted for gender, age, the purpose of flight, language, other health behavior and all other Socio-economic variables (occupation, income, education, marital status).

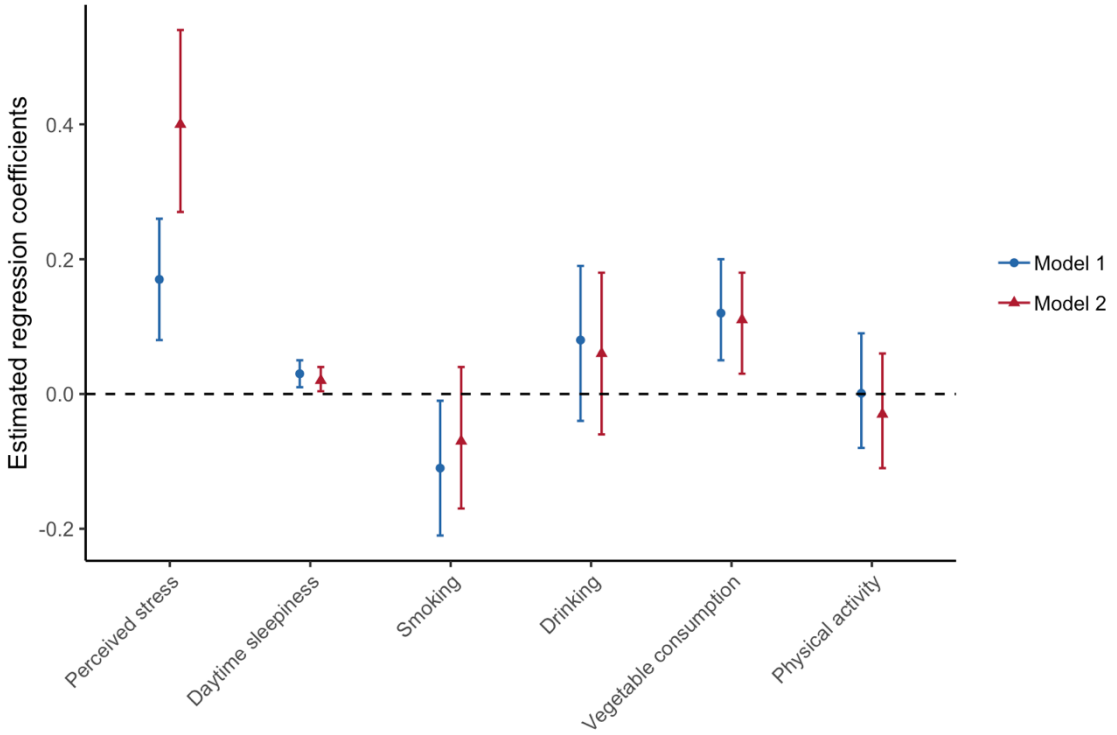

Figure 4. Estimated regression coefficients of jet lag symptom on health status / behavior. Model 1: Adjusted for gender, age, the purpose of flight, language and other health behavior. Model 2: Adjusted for gender, age, the purpose of flight, language, other health behavior and all other Socio-economic variables (occupation, income, education, marital status).

The results of multivariate linear regression analysis applying Model 2 as stated above are briefly summarized in Table 2. A positive association was found between jet lag perception and perceived stress, but a negative association was seen between jet lag perception and smoking. Positive associations were found between the number of jet lag symptoms and perceived stress, daytime sleepiness, and vegetable consumption.

Table 2. Overall findings between lifestyle factors and jet lag perception and symptoms

	Jet lag perception	Jet lag symptoms
	Association[a]	Association[a]
Perceived stress		
Model 2[b]	++	+++
Daytime sleepiness		

Model 2[b]		+
Smoking		
Model 2[b]	--	
Drinking		
Model 2[b]		
Vegetable consumption		
Model 2[b]		++
Physical activity		
Model 2[b]		

[a] + means positive association, while – means negative association.
+++ or ---: $p<0.001$, ++ or --: $p<0.01$, + or -: $p<0.05$
[b] Model 2: Adjusted for gender, age, the purpose of flight, language, other health behavior and all other Socio-economic variables (occupation, income, education, marital status).

Discussion

In the present study, we found that several lifestyle factors are associated with jet lag perception or symptoms. Previous studies have found that stress and drinking behavior (excessive caffeine and alcohol intake) are related to both jet lag perception and symptoms because these factors affect the quality of sleep [11]. Our study's results partially support these previous findings.

In terms of stress, our study found that higher perceived stress level is associated with more severe jet lag perception and higher number of jet lag symptoms. This is consistent with previous research. There is considerable evidence that stress and circadian rhythms are linked [22]. Our findings are supported by the fact that jet lag is known to be caused by a mismatch between circadian rhythms and the external environment.

There is existing evidence of a relationship between sleep deprivation and jet lag [8], which our findings support. People experiencing daytime sleepiness also reported a higher number of jet lag symptoms, while it was not associated with jet lag perception. We assume that people who have daytime sleepiness may have experienced some sleep deprivation symptoms, and may already be familiar with them. Those people may

therefore not perceive their jet lag as severe. However, as they declared higher numbers of jet lag symptoms than those who do not have daytime sleepiness, they did tend to experience severe jet lag. Jet lag may bring additional difficulty for usual daytime sleepiness and also these people do not perceive jet lag severely, those who have usual sleep issues may require extra care and prevention of jet lag.

Contrary to our hypotheses, an inverse association was found between smoking and jet lag perception. In other words, smokers reported less severe perceptions of jet lag. Some studies have found that smoking affects circadian rhythms [23], and there is considerable evidence that smoking causes sleep problems [24]. Therefore, we assume that smokers may already have had bad sleep quality in their daily life, and may not perceive jet lag as severe. Also, there is evidence that smokers tend to have an optimistic bias and unrealistic beliefs in terms of health risk assessment and their behaviors [25]. Although previous research on the subject does not mention jet lag perception specifically, considering these previous findings, it is possible that smokers' optimistic bias contributes to perceiving jet lag less seriously. In fact, there was no association between smoking behavior and the number of jet lag symptoms.

Physical activity was also negatively associated with jet lag perception. As in research on smoking, some studies have found a relationship between physical activity and optimism [26]. These qualities could lead to lower jet lag perception. Although previous studies have seen a relationship between alcohol intake and jet lag [27], we did not find any association between general alcohol intake and jet lag (perception or the number of symptoms). In our study, we asked the general alcohol intake. The response might have been different if we had specifically asked about alcohol intake at pre-, during, and post-flight.

Last but not least, our study suggests that people reporting higher vegetable intake are more likely to report a higher number of jet lag symptoms. This is also contrary to what we expected. We consider that people having higher vegetable consumption are a more health-conscious population. It means that they may be more cautious about jet lag symptoms (e.g., due to their knowledge and/or understanding of jet lag symptoms, etc). Or, these people may have already had some health issues, which could explain their more frequent vegetable intake.

Limitations

There are several limitations in this study. First, since this is a cross-sectional study, we can demonstrate only associations. Estimated coefficients could be biased due to endogeneity (e.g., jet lag perception and perceived stress could affect each other, resulting in simultaneity of these variables). Second, recall bias could be a factor. Our recruiting criteria limited our sample to people who had business class experience flying within the past year. Thus, their responses could have been affected by the timing of their flights. Also, there is a lack of generalizability, and these findings may not be applicable for the general population. Third, jet lag was only measured at one time. In addition, we asked participants about their general experience of flying, not of one particular flight. Since jet lag has individual variation due to other factors such as their travel direction and length [8], responses may have been affected. In order to increase accuracy in reporting, it would be better to measure jet lags multiple times, and also multiple flight experiences within one person. Another limitation is due to our asking about general alcohol intake, and not asking specifically about alcohol intake during a flight. There are data suggesting that air passengers tend to increase their alcohol intake during flights [28], so our result could change if questions were asked differently. Last but not least, although multiple members of the project checked the translation of the questions about jet lag perception, we have not validated whether these Japanese and English versions of the survey are compatible.

Despite these limitations, there have been few studies about the determinants of jet lag, and our findings help develop further interventions for jet lag, especially in terms of prevention. Finding lifestyle factors for jet lag makes it possible to offer evidence-based jet lag prevention strategies for travelers and flight crews before, during, and after flights.

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